Robot Localization and Mapping - Homework 4

Tianzhi Li
tianzhil@andrew.cmu.edu

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1 Overview

2 Iterative Closest Point (ICP)

2.1 Projective data association

The condition is

$$0 \leq u < W; 0 \leq v < H; 0 \leq d$$

This step is necessary because projective correspondences are found based on 2D projections, which may not accurately reflect true 3D spatial relationships. As a result, the matched point q from the vertex map may be far from the original point p in 3D space or lie on a different surface, leading to incorrect associations. To ensure reliable correspondences, additional filtering using a distance threshold (to enforce spatial closeness) and an angle threshold (to ensure surface normal consistency) is required. This helps reject mismatches and improves the robustness of point cloud registration.

2.2 Linearization

We are given the residual:

$$r_i^2(\delta R, \delta t) = \left\| n_{q_i}^\top \left((\delta R) p_i' + \delta t - q_i \right) \right\|^2$$

where

$$p_i' = R^0 p_i + t^0$$

Using the small-angle approximation, we parameterize the rotation as:

$$\delta R \approx I + \begin{bmatrix} 0 & -\gamma & \beta \\ \gamma & 0 & -\alpha \\ -\beta & \alpha & 0 \end{bmatrix} = I + [\omega]_\times \quad \text{where} \quad \omega = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

Thus:

$$\delta R\, p_i' \approx p_i' + [\omega]_\times p_i'$$

And the cross product has the identity:

$$[\omega]_\times p_i' = -[p_i']_\times \omega$$

Thus:

$$\delta R\, p_i' \approx p_i' - [p_i']_\times \omega$$

Substituting back into the residual expression:

$$r_i(\omega, \delta t) = n_{q_i}^\top (p_i' - [p_i']_\times \omega + \delta t - q_i)$$

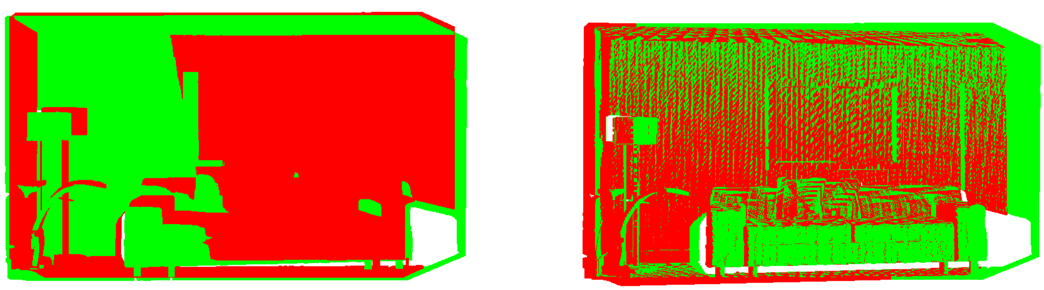
Let:

$$A_i = \begin{bmatrix} -n_{q_i}^\top [p_i']_\times & n_{q_i}^\top \end{bmatrix}, \quad b_i = n_{q_i}^\top (p_i' - q_i)$$

Then the residual becomes:

$$r_i(x) = A_i x + b_i$$

2.3 Optimization



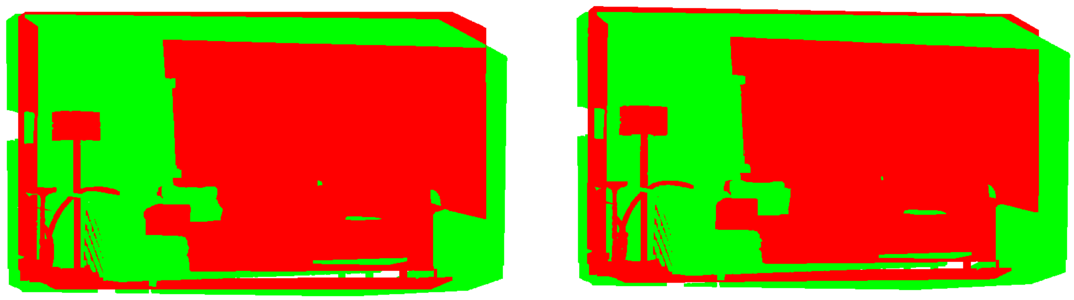
(a) Before

(b) After

Figure 1: Point clouds before and after registration (frame 10 and 50)

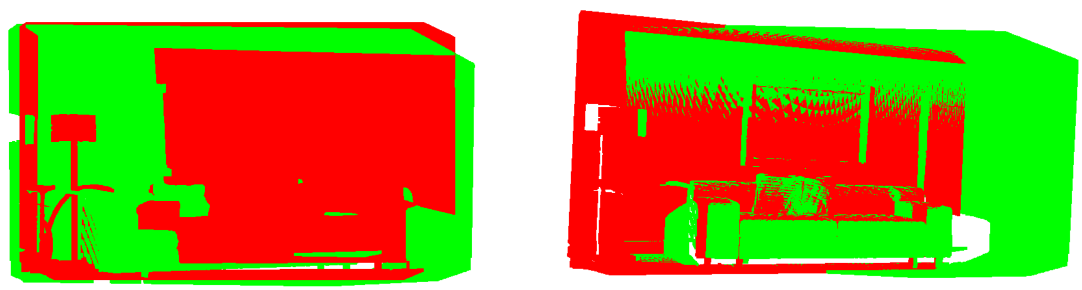
We first applied ICP between frame 10 (source) and frame 50 (target). Before ICP, the point clouds are visibly misaligned. After running ICP with the default settings, the alignment significantly improves, showing successful convergence (the loss is $8.89e^{-6}$ and inlier count is 281370 after 10 iterations). This is expected since the motion between the frames is small and well-suited for ICPs assumptions.

Next, we tested a more challenging case: frame 10 to frame 100. The initial misalignment is much larger, and early ICP iterations show minimal loss reduction (the loss is $1.28e^{-3}$ and the inlier count is 28525 after 10 iterations), which indicats poor correspondence and slow



(a) Before (b) After

Figure 2: Point clouds before and after registration (frame 10 and 100) with `iteration` = 10



(a) Before (b) After

Figure 3: Point clouds before and after registration (frame 10 and 100) with `iteration` = 100

convergence. However, by increasing the iteration count to 100, ICP eventually converges with a low final loss ($5.56e^{-5}$) and high inlier count (231627), resulting in good alignment.

This experiment shows that while ICP can succeed with large frame gaps, it is highly sensitive to the initial pose. Without a good initial guess or enough overlap, convergence is slow and unreliable unless more iterations are used.

3 Point-based Fusion

3.1 Filter

Functions `filter_pass1` and `filter_pass2` are implemented in `fusion.py`.

3.2 Merge

The weighted average of the position is:

$$\bar{p} = \frac{w \cdot p + (R_c^w q + t_c^w)}{w+1}$$

The weighted average of normals is (before normalization):

$$\bar{n} = \frac{w \cdot n_p + R_c^w n_q}{w + 1}$$

Then normalize the result:

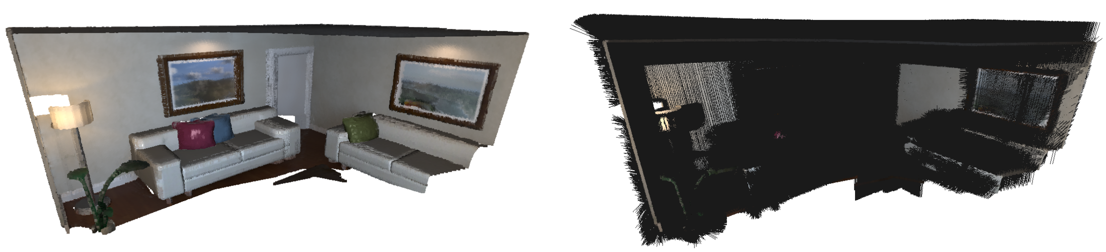
$$\bar{n}_{\text{normalized}} = \frac{\bar{n}}{\|\bar{n}\|}$$

Function `merge` is implemented in `fusion.py`.

3.3 Addition

Function `add` is implemented in `fusion.py`.

3.4 Results



(a) Color map

(b) Normal map

Figure 4: Fusion with ground truth poses with a normal map

The number of points in the final map is 1362218. If I naively concatenate all the input, the number of points will be $200 \times \frac{480}{2} \times \frac{640}{2} = 15,360,000$. The compression ratio is 0.0887.

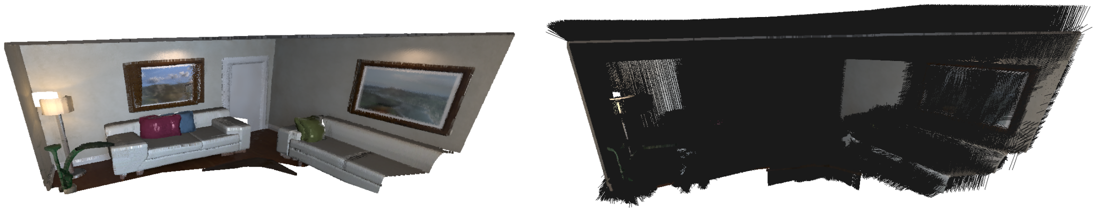
4 The dense SLAM system

The source is the input RGBD-frame and the target is the map. This is because ICP tries to align the source to the target. We're aligning the new frame to the existing map, so the map acts as a fixed reference.

I don't think their roles can be swapped. Although ICP is symmetric in formulation, but in practice, it's hard to regard the map as the source. There are two reasons:

1. The source is actively transformed to match the target. Swapping roles would mean you're trying to fit the entire map to a single RGBD frame, which doesn't make sense in a mapping context.

2. Also, the vertex and normal maps are often generated from the target (map), so the implementation assumes a fixed target and a moving source.



(a) Color map (b) Normal map

Figure 5: Fusion with poses estimated from frame-to-model ICP

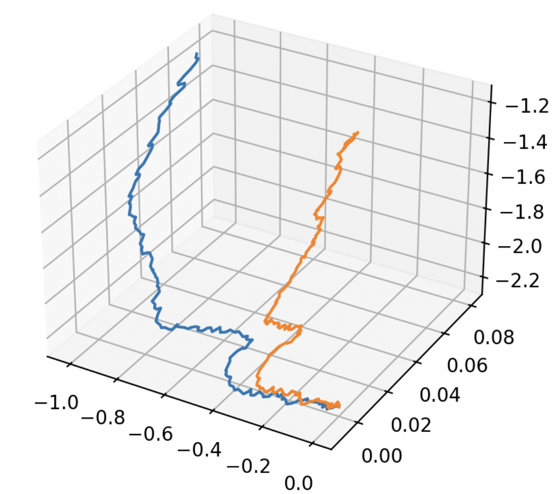


Figure 6: Trajectory

The visualization is shown in Figure 5.
The estimated trajectory and the ground truth is shown in Figure 6.